

Cognitive Dimensions: A Unified Learning Framework

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NOTICE: This paper presents the **5 Cognitive Dimensions (5CD)** theory, serving as a foundational basis for discussion, hypothesis generation, and future empirical research. While the framework is grounded in established principles of cognitive science, its validity has not yet undergone rigorous evaluation. **This should not be treated as validated diagnostic conclusions** at this stage. The author welcomes critical feedback and proposals for collaborative investigation to refine these constructs further.

Abstract

Understanding a learner's unique cognitive style is the ultimate goal of personalized development, yet current models struggle to adequately portray the complexity of the underlying cognitive mechanisms operating across diverse cognitive styles. This paper introduces a unified framework that fills in the gaps in Kolb's experiential learning theory. We propose a new taxonomy based on five spectrums of continuous dimensions: Activation, Engagement, Regulation, Acquisition, and Mapping. These bridge the gap between internal mental processes and external learning stimuli. We also present a modified Likert-scale protocol that eliminates self-reporting bias. This approach allows organizations to accurately map learner archetypes, resulting in more personalized, data-driven training and development strategies.

1. Introduction

The central challenge in educational psychology is accommodating the wide diversity of individual differences in learning rates, approaches, and needs. Many early attempts to address this challenge relied on sensory-based models, such as the visual–auditory–read/write–kinesthetic (VARK) model developed by Neil Fleming in 1987. The term *learning styles* refers to the idea that individuals differ in which mode of instruction or study is most effective for them (Pashler et al., 2008). However, extensive empirical reviews have concluded that there is no credible evidence supporting the hypothesis that matching instructional methods to a learner's preferred style reliably improves learning outcomes (Newton & Miah, 2017; Zrudlo, 2023).

In contrast, research from cognitive science has produced well-supported theories with clear implications for learning and memory (Regehr & Norman, 1996). The acquisition of expertise can be characterized by the development of idiosyncratic memory structures, often described as semantic networks—meaningful patterns of connections among abstract concepts and specific experiences. Information, such as the assumptions and hypotheses required for diagnosis or decision-making, is retrieved through the activation of these networks.

A wide range of variables influences the capacity to store and retrieve information from memory, including meaning, context, the manner in which information is learned, and the extent of relevant retrieval practice. Much of expertise development involves a transition from reliance on general problem-solving strategies to the use of specialized knowledge that reduces the need for explicit problem-solving. Two key manifestations of this specialization are the use of analogy and the adaptation of general cognitive routines to specific domains. Experts possess both abstract, prototypical representations of categories and a rich collection of concrete examples derived from experience. These complementary forms of knowledge are jointly employed in categorization and diagnostic judgment.

Together, these findings indicate that learning is critically shaped by internal cognitive processes, rather than being determined solely by external learning conditions. However, existing educational models often fail to integrate these two dimensions in a coherent way. This paper proposes a conceptual bridge grounded in established theories

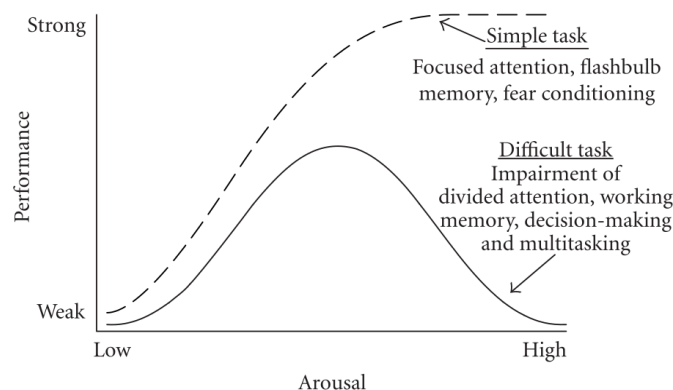
of cognitive styles, which is later formalized as a new taxonomy for understanding learning and problem-solving differences.

1.1 Cognitive Style Models

Cognitive style is a concept in cognitive psychology used to describe stable differences in how individuals perceive, process, organize, and remember information. Rather than measuring ability or intelligence, cognitive styles reflect **preferences in thinking and learning**. Several well-established psychological theories provide foundational mechanisms that inform the model proposed in this paper.

One of the earliest relevant principles is the **Yerkes–Dodson law**, formulated by Robert Yerkes and John Dodson in 1908. Based on early experimental research using animal models and controlled stimulus conditions, the law describes the relationship between arousal and performance (Yerkes & Dodson, 1908). It states that performance improves as mental or physiological arousal increases, but only up to an optimal point, beyond which further arousal leads to performance decline. Mental arousal may include states such as alertness or excitement. This relationship is commonly illustrated as an inverted U-shaped curve.

Figure 1: Yerkes–Dodson law (Yerkes & Dodson, 1908)



Building on arousal-based accounts of performance such as the Yerkes–Dodson law, **Marvin Zuckerman** introduced the construct of **sensation seeking** to explain stable individual differences in preferred levels of stimulation (Zuckerman, 1979). Sensation seeking is defined as the tendency to seek out novel, complex, and intense experiences, as well as the willingness to tolerate physical, social, or cognitive risk in pursuit of such experiences. According to Zuckerman's **Optimal Stimulation Theory**, individuals differ in their baseline arousal levels and therefore actively regulate their environments to achieve an optimal level of stimulation. Individuals high in sensation seeking tend to require higher levels of arousal to remain engaged and perform optimally, whereas individuals low in sensation seeking prefer calmer, more predictable conditions and may experience performance decrements under excessive stimulation.

In the 1970s, **Gordon Pask** introduced *Conversation Theory* (CT), a cybernetic and constructivist framework describing how knowledge is formed and shared through interaction (Scott, 2001). Unlike models that treat learning as the passive transmission of information, Pask viewed learning as an active, dialogical process in which participants iteratively adjust their understanding by “steering” one another’s perspectives toward mutual comprehension.

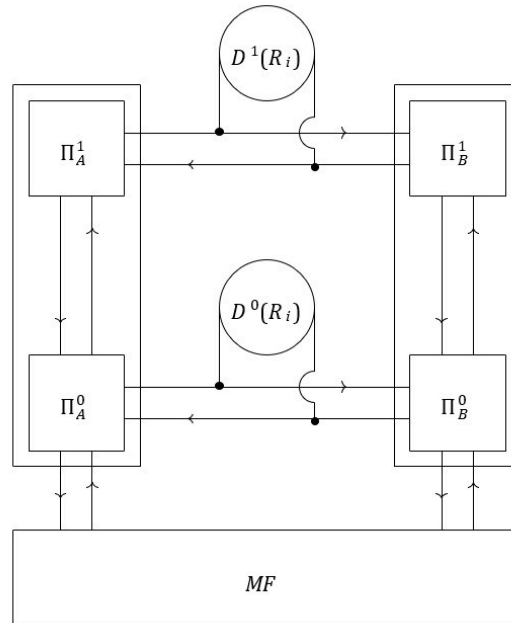
Within this framework, Pask identified two primary cognitive styles that describe how individuals engage with conceptual structures:

- **Serialists**, who prefer a sequential, step-by-step progression through ideas, and

- **Holists**, who seek overarching patterns and relationships before attending to details in ideas.

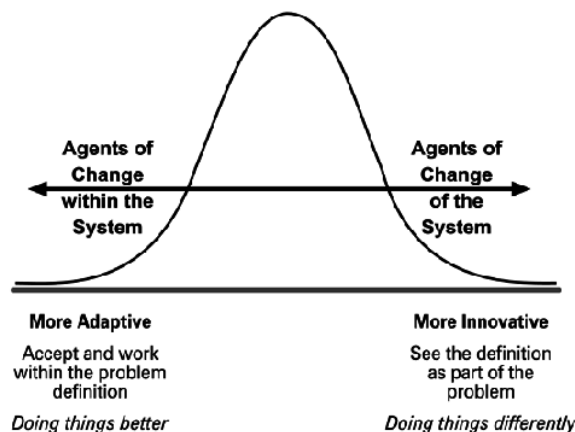
Figure 2: Example of a conversational skeleton between A and B.

Each participant contains procedural repertoires Π that modulate the input and output, speech-acts, or descriptions of topic relations $D(R_i)$, and a modelling facility MF . (Pask, 1975)



Another influential cognitive style model was proposed by **Michael Kirton** in his *Kirton Adaptation-Innovation Inventory (KAI) theory* (1976). Kirton argued that individuals differ in their preferred approaches to problem solving along a continuum ranging from high adaptation to high innovation (Stum, 2009). *Adaptors* tend to focus on improving existing systems, valuing structure, efficiency, and refinement, whereas *innovators* are more inclined to challenge assumptions and redefine problems, often producing more radical change. Kirton also developed the KAI to measure these preferences, demonstrating their relevance across a wide range of organizational and creative contexts.

Figure 3: The Adaptation-Innovation continuum of cognitive style (Jablokow et al., 2011)



A major line of research in cognitive psychology distinguishes between two modes of information processing, commonly referred to as **dual-process theories** (Evans, 2008; Kahneman, 2011). These theories propose that human cognition operates along a continuum between two qualitatively different types of processing.

The first mode, often called *Type 1 processing* (or System 1), is fast, automatic, intuitive, and requires little conscious effort. It relies heavily on heuristics, pattern recognition, and prior experience, enabling individuals to make quick judgments with minimal cognitive load. While efficient, this mode is more susceptible to biases and errors, particularly in complex or novel situations.

The second mode, known as *Type 2 processing* (or System 2), is slow, deliberate, analytical, and effortful. It requires conscious attention and working memory resources, and it is responsible for abstract reasoning, logical evaluation, and the correction of intuitive responses. Type 2 processing enables individuals to override initial intuitions and engage in deeper analysis when accuracy and understanding are prioritized.

Importantly, individuals differ in their propensity to engage in Type 2 processing, even when cognitive ability is held constant. This dispositional tendency is captured by the construct **Need for Cognition (NFC)**, defined as an individual's tendency to enjoy and engage in effortful cognitive activities (Cacioppo & Petty, 1982). Individuals high in NFC are more likely to allocate cognitive resources to analytical processing, scrutinize arguments, and persist in complex reasoning tasks, whereas individuals low in NFC tend to rely more on intuitive or heuristic processing strategies.

Empirical work has further operationalized this distinction through the **Cognitive Reflection Test (CRT)**, which measures an individual's tendency to inhibit an immediate intuitive response in favor of reflective reasoning (Frederick, 2005). Performance on the CRT has been shown to relate to NFC, suggesting that motivational disposition and reflective engagement jointly influence the depth of cognitive processing (Toplak, West, & Stanovich, 2011).

In the late 1980s, **John Sweller** developed **Cognitive Load Theory (CLT)**, grounded in the architecture of human cognition (Sweller, 2011). CLT posits that working memory has a limited capacity, while long-term memory is effectively unlimited and stores knowledge in organized structures known as *schemas*. Learning, from this perspective, involves the construction and refinement of schemas, which enable information to be chunked and processed more efficiently, thereby reducing the burden on working memory. Effective instructional design seeks to manage cognitive load so that mental resources are allocated toward learning rather than unnecessary effort.

Sweller identified three types of cognitive load that influence learning outcomes:

1. **Intrinsic load**, which reflects the inherent complexity of the material and cannot be eliminated, though it can be managed through sequencing and scaffolding strategies.
2. **Extraneous load**, which arises from poorly designed instruction and imposes unnecessary cognitive burden; and
3. **Germane load**, which represents the productive mental effort devoted to schema construction and knowledge integration.

Two earlier constructivist theorists, **Jean Piaget** and **Lev Vygotsky**, also made foundational contributions to understanding learning and cognition (Tudge & Winterhoff, 1993). Piaget's theory of cognitive development emphasized the individual construction of knowledge through interaction with the physical environment, whereas Vygotsky's sociocultural theory highlighted the role of social interaction, language, and cultural tools in shaping cognitive development. Together, their work underscores the interplay between individual cognition and external learning contexts.

Numerous other robust cognitive and educational theories could be considered. However, as will be demonstrated in the following sections, the models reviewed above are sufficient to support the development of a unified framework that integrates cognitive style, learning processes, and motivational dynamics.

We now turn to the organization of these previously isolated variables into a single integrated model consisting of five dimensions. This synthesis begins with an evolution of experiential learning theory.

Figure 4: Piaget's Theory vs Vygotsky's Theory (Tudge & Winterhoff, 1993)

	Piaget	Vygotsky
Sociocultural context	Little emphasis	Strong emphasis
Constructivism	Cognitive constructivist	Social constructivist
Stages	Strong emphasis on stages of development	No general stages of development proposed
Key processes in development & learning	Equilibration; schema; adaptation; assimilation; accommodation	Zone of proximal development; scaffolding; language/dialogue; tools of the culture
Role of language	Minimal – Language provides labels for children's experiences (egocentric speech)	Major – Language plays a powerful role in shaping thought
Teaching implications	Support children to explore their world and discover knowledge	Establish opportunities for children to learn with the teacher and more skilled peers

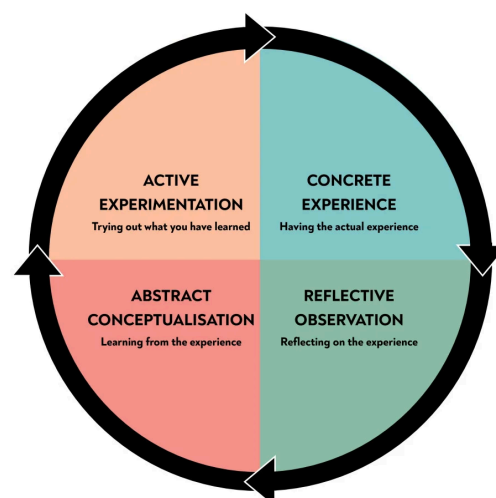
1.2 Evolving the Experiential Model

In 1984, **David A. Kolb** introduced *Experiential Learning Theory* (ELT), from which he later developed the Learning Styles Inventory (Kolb, 1984). ELT operates on two related levels: a four-stage learning cycle that describes how learning unfolds through experience, and a set of learning style preferences derived from this cycle. Kolb famously defined learning as “the process whereby knowledge is created through the transformation of experience,” emphasizing learning as an active, ongoing process rather than a passive accumulation of information.

The ELT learning cycle consists of four stages through which learners may move repeatedly:

1. **Concrete experience (CE)**: encountering a new situation or reinterpreting an existing experience through new ideas;
2. **Reflective observation (RO)**: reflecting on the experience in relation to prior knowledge;
3. **Abstract conceptualization (AC)**: forming new concepts or modifying existing ones based on reflection;
4. **Active experimentation (AE)**: applying these concepts in practice to test their implications.

Figure 5: Experiential Learning Cycle



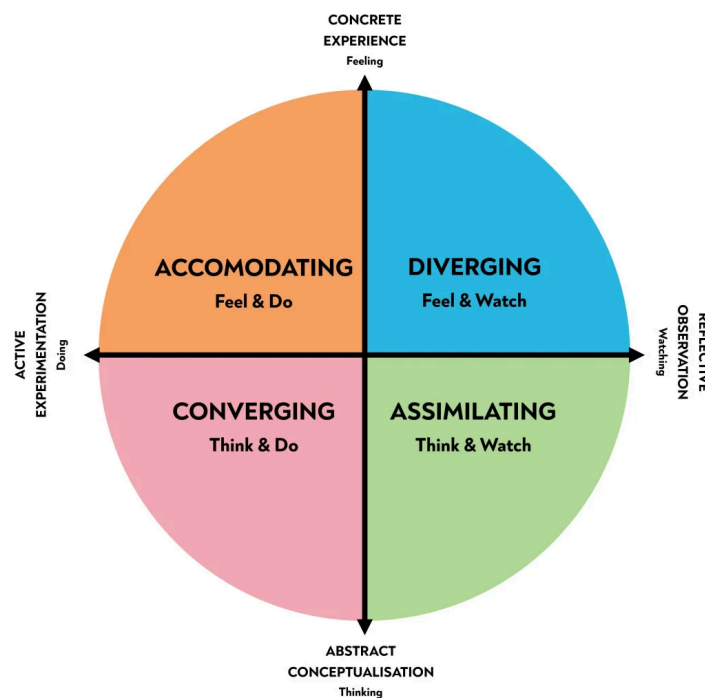
According to Kolb, effective learning involves engaging with all four stages, although individuals tend to develop preferences for certain modes over others. These preferences give rise to distinct learning styles, which Kolb conceptualized using two intersecting continua. The **Perception Continuum** describes how individuals prefer to take in information, ranging from concrete experience to abstract conceptualization, while the **Processing**

Continuum describes how they prefer to act on information, ranging from reflective observation to active experimentation.

From the interaction of these two dimensions, Kolb identified four learning styles:

- **Diverging (CE/RO)** — characterized by viewing situations from multiple perspectives and emphasizing observation over action. Individuals with this style tend to be imaginative, emotionally aware, open to feedback, and interested in people and cultural contexts.
- **Assimilating (AC/RO)** — marked by a preference for clear explanations, logical structure, and abstract concepts. These learners value theoretical coherence and analytic models more than social interaction.
- **Converging (AC/AE)** — oriented toward problem solving and practical application of ideas. Individuals with this style favor technical tasks, experimentation, and the use of concepts to arrive at concrete solutions.
- **Accommodating (CE/AE)** — characterized by a hands-on, intuitive approach to learning. These learners prefer action, adapt quickly to new challenges, and often rely on trial and error rather than extensive analysis.

Figure 6: Kolb's Learning Styles Matrix



To operationalize ELT, Kolb developed the **Learning Styles Inventory (LSI)**, a self-assessment tool that typically requires 30–45 minutes to complete. The LSI classifies individuals into one of the four learning styles based on their relative preferences along the perception and processing continua. While widely used in educational and organizational settings, the model has also prompted ongoing discussion regarding the stability and interpretation of learning style classifications.

While Experiential Learning Theory (ELT) has been highly influential in education and management development, it has also attracted substantial criticism regarding both its theoretical foundations and practical applications (Morris, 2020). Several key limitations have been identified in the literature.

First, ELT has been criticized for its weak theoretical and empirical support. Multiple scholars argue that the model lacks strong empirical validation and that Kolb's interpretation of earlier foundational theorists such as Dewey and Lewin oversimplifies or misrepresents their original ideas (Jarvis, 1987; Coffield et al., 2004; Miettinen, 2000).

Second, the model has been described as an oversimplification of learning processes. Learning rarely follows a clean, sequential cycle, and rigidly categorizing learners into four styles risks limiting flexibility and misrepresenting how learning unfolds in real-world contexts.

Third, ELT has been faulted for its limited consideration of social, cultural, and emotional factors. The theory emphasizes individual cognition while underplaying the influence of social interaction and cultural context, and its associated inventory has been applied primarily within Western populations.

Finally, methodological concerns have been raised about the Learning Styles Inventory (LSI). As a self-report instrument, it reflects perceived rather than observed behavior, and research has shown that matching instruction to ELT-based learning styles does not reliably improve learning outcomes.

In light of these limitations, this paper proposes a new integrative perspective that seeks to address the gaps left by ELT. Rather than discarding experiential learning, the proposed framework synthesizes ELT with established cognitive style models to capture both the cyclical nature of experience and the psychological, emotional, and social factors that shape learning. This integration is summarized by the following general relationship:

$$\textit{learning} = \textit{internal traits} \times \textit{external conditions}$$

This formulation emphasizes that learning emerges from the interaction between relatively stable individual characteristics and situational factors, rather than from either dimension alone.

2. Five Cognitive Dimensions

To operationalize the interaction between internal traits and external conditions, we introduce a five-dimensional framework that organizes these factors into a coherent cognitive taxonomy.

Learning can be understood as the process of adapting to new knowledge, behaviors, skills, or values. This process is driven by the interaction of internal traits and external conditions. Internal traits include relatively stable individual characteristics such as cognitive preferences, psychological tendencies, biological factors, and prior knowledge. External conditions include elements of the learning environment, such as physical settings, social and cultural influences, instructional design, and available resources.

We synthesize these learning-related factors into five measurable cognitive dimensions. Each dimension represents a continuous spectrum along which individuals may vary. We propose that these dimensions are orthogonal, capturing distinct aspects of cognition without redundancy. Together, they describe how individuals engage with learning experiences and form what we refer to as a learner's cognitive style and process. This framework is not intended to measure cognitive ability, such as intelligence quotient (IQ), memory capacity, or attentional limits, which primarily influence learning speed or depth rather than learning approach.

Starting from this point, we refer to this framework as the **Five Cognitive Dimensions (5CD)** theory.

Figure 7: Five Cognitive Dimensions



2.1 Dimension 1: Activation

The first dimension describes how much **cognitive activation** and **environmental stimulation** an individual typically relies on for optimal engagement and learning. More precisely, it measures the degree of stimulation, pressure, novelty, or intensity a person requires to enter and sustain an effective cognitive state.

This dimension is motivated by the Yerkes–Dodson law, which demonstrates that performance varies as a function of arousal, and more importantly, by Zuckerman’s Optimal Stimulation Theory, which explains stable individual differences in required levels of stimulation. While the Yerkes–Dodson law describes a general performance pattern, Zuckerman’s work accounts for why the level of arousal that facilitates learning and performance differs across individuals. Together, these theories support the notion that learners vary in how much stimulation they require to engage effectively, independent of motivation or cognitive ability.

We refer to this dimension as **Activation**, with the two endpoints of the spectrum labeled **Grounded** and **Energized**.

- **Grounded** represents a reliance on low levels of stimulation when learning or solving problems. Individuals who lean toward grounded activation tend to engage most effectively in calm, predictable, and low-pressure environments. They are more sensitive to overstimulation and may experience reduced performance under excessive urgency, noise, or rapid change. This orientation favors psychological safety, sustained focus, and careful, deliberate work.
- **Energized** represents a reliance on higher levels of stimulation to achieve engagement. Individuals who lean toward energized activation tend to perform best under conditions of novelty, urgency, and challenge. They may become under-stimulated in routine or low-pressure contexts and often seek dynamic environments to maintain cognitive engagement. This orientation supports rapid mobilization, adaptability, and effective functioning in high-intensity or fast-changing situations.

Figure 8: First Dimension



2.2 Dimension 2: Engagement

The second dimension describes how deeply an individual tends to process information before reaching understanding, judgment, or action. More precisely, it measures **cognitive engagement depth**, ranging from satisficing through intuitive, low-effort processing to elaborating through sustained, reflective reasoning.

This dimension is motivated by Need for Cognition (NFC) theory, as well as research on dual-process theories of cognition and the Cognitive Reflection Test (CRT). NFC captures stable individual differences in the tendency to enjoy and engage in effortful cognitive activities, while CRT operationalizes the ability to inhibit an initial intuitive response in favor of reflective reasoning. Together, these constructs distinguish individuals not by cognitive capacity, but by their willingness to engage in deeper cognitive processing when it is available as an option.

We refer to this dimension as **Engagement**, with the two endpoints of the spectrum labeled **Satisficing** and **Elaborating**.

- **Satisficing** represents a tendency for efficient, intuitive processing that reaches a “good enough” conclusion with minimal cognitive effort. Individuals who lean toward satisficing rely more on heuristics, surface cues, and prior patterns, terminating cognitive processing once sufficient confidence is achieved. This approach favors speed, fluency, and cognitive economy, and is often adaptive in low-stakes, time-constrained, or familiar contexts.
- **Elaborating** represents a tendency for deep, reflective engagement with information. Individuals who lean toward elaborating tend to invest sustained cognitive effort, actively evaluate assumptions, and seek coherence and justification before forming conclusions. They are more tolerant of ambiguity and delay closure to explore alternatives, verify reasoning, and integrate new information into existing mental models.

Figure 9: Second Dimension



2.3 Dimension 3: Regulation

The third dimension describes what keeps an individual’s thinking “on track” by default while learning—whether regulation primarily occurs within the individual through self-monitoring and internal feedback, or externally through social interaction, guidance, and environmental scaffolds. More precisely, it measures where cognitive control originates during learning and problem-solving.

This dimension is inspired by developmental and sociocultural theories of learning, most notably Piaget’s account of self-regulated cognitive development and Vygotsky’s sociocultural theory, which emphasizes externally mediated learning through dialogue, guidance, and cultural tools. Together, these perspectives highlight that individuals differ not only in what they learn or how deeply they process information, but also in where the regulatory feedback that guides understanding and correction primarily comes from.

We refer to this dimension as **Regulation**, with the two endpoints of the spectrum labeled **Internal regulation** and **External regulation**.

- **Internal regulation** describes a learning style in which individuals primarily guide their own thinking through self-reflection, internal feedback, and personal error monitoring. Internally regulated learners tend to evaluate their understanding independently, adjust strategies on their own, and maintain cognitive direction without requiring frequent external input. This mode supports autonomy, self-directed learning, and the development of strong internal models of correctness and progress.
- **External regulation** describes a learning style in which individuals rely more heavily on feedback, guidance, and structure provided by others or by the environment. Externally regulated learners benefit from dialogue, examples, prompts, and scaffolding that help orient their thinking and correct misunderstandings. This mode supports learning through social interaction, shared meaning-making, and the use of external cues to stabilize understanding, particularly in complex or unfamiliar domains.

Figure 11: Third Dimension



2.4 Dimension 4: Acquisition

The fourth dimension describes how experience is first transformed into understanding when an individual encounters new information or a novel task. More precisely, it measures the primary route through which knowledge is initially grounded—either through direct interaction and action, or through observation and mental modeling before acting.

This dimension should not be interpreted as a sensory learning style or as a claim that learning outcomes improve when instruction is matched to an individual's position on the spectrum. Rather, it captures how experience is initially anchored into understanding. Both modes are necessary for developing expertise, and effective learning often involves movement across the spectrum depending on task demands and experience level.

This dimension refines the experiential emphasis of Kolb's learning theory by separating experience acquisition from prescriptive learning cycles. It is supported by research in embodied and enactive cognition, distinctions between procedural and declarative knowledge, and instructional findings on observation-based learning and problem solving. Together, these perspectives suggest that individuals differ in how experience is first grounded, not in what method works best for learning.

We refer to this dimension as **Acquisition**, with the two endpoints of the spectrum labeled **Observational** and **Enactive**.

- **Observational** refers to grounding understanding through watching, modeling, and mental simulation. Observational learners tend to form internal representations before acting, allowing them to anticipate outcomes and reduce uncertainty when engaging with new tasks.
- **Enactive** refers to grounding understanding through direct action and interaction. Enactive learners tend to explore tasks by doing, using immediate feedback from the environment to form meaning and build practical understanding.

Figure 12: Fourth Dimension



2.5 Dimension 5: Mapping

The fifth dimension describes how individuals relate to **structure** when thinking, learning, and solving problems. More precisely, it measures the degree to which a person tends to maintain existing cognitive structures or transform them when dealing with complexity.

This dimension synthesizes insights from Pask's Conversation Theory, which distinguishes between serialist and holist learners, and Kirton's Adaptation-Innovation theory, which places problem-solving aptness along a continuum from adaptation to innovation. In this synthesis, a *serialist-adaptor* tends to favor sequential accumulation of knowledge and the preservation of existing structures, whereas a *holist-innovator* tends to engage in pattern-level reasoning and the transformation of structure.

We refer to this dimension as **Mapping**, with the two endpoints of the spectrum labeled **Building** and **Reframing**.

- **Building** is a way of thinking that focuses on improving what already exists. Builders usually accept a problem as it is given and work step by step to make it better. They need clear structure, stable rules, and gradual progress, adding new pieces carefully so the overall system stays solid. This style of thinking creates value through refinement, optimization, and reliability, often resulting in solutions that are practical, durable, and easy for others to follow.
- **Reframing** is a way of thinking that focuses on changing how a problem is understood. Reframers tend to question assumptions, look for broader patterns, and explore different perspectives before settling on a solution. They are comfortable with uncertainty and often jump between ideas to form a new overall picture. This approach can reveal possibilities that were hidden before and often leads to novel insights or breakthrough solutions.

Figure 10: Fifth Dimension



3. Thirty-Two Cognitive Species

The Five Cognitive Dimensions (5CD) model could be summarized in the following table.

Table 1: Five Cognitive Dimensions

No.	Dimension	Description	Left Endpoint	Right Endpoint
1.	Activation	The level of stimulation and intensity required for effective learning and thinking	Grounded	Energized
2.	Engagement	How deeply an individual processes information before forming an understanding or acting	Satisficing	Elaborating
3.	Regulation	Where cognitive control and corrective feedback primarily come from during learning	Internal	External
4.	Acquisition	How experience is first converted into understanding	Observational	Enactive
5.	Mapping	How an individual relates to structure when reasoning and solving problems	Building	Reframing

Note that these dimensions describe tendencies, not fixed traits. They are orthogonal, where each captures a distinct cognitive mechanism. They do not imply learning effectiveness through “matching”. Effective learning often involves movement across dimensions depending on task demands, context, and expertise.

Some dimensions determine how learning is activated and regulated, while others determine how knowledge itself is constructed and represented. Therefore, we could naturally categorize the 5CD into two categories. The first three—Activation, Engagement, and Regulation—determine whether and how thinking is activated and sustained. We call this category **State Group**. The last two—Acquisition and Mapping—determine how knowledge is structured and grounded. We call this category **Structure Group**.

Although each parameter is modelled in a continuous spectrum, we are interested in the case when an individual shows a dominant trait towards each one of the endpoints in each dimension. Considering the extreme cases, each of the five dimensions has two possible states, so the total possible state combinations are:

$$\binom{5}{0} + \binom{5}{1} + \binom{5}{2} + \binom{5}{3} + \binom{5}{4} + \binom{5}{5} = 2^5 = 32.$$

So we have 32 possible combinations of the endpoint arrangements. We refer to these characteristics as the **Cognitive Species**. These cognitive species represent configurations of tendencies across five independent dimensions rather than fixed categories or personality types. Individuals may shift between species across contexts, tasks, and developmental stages. We will discuss these species within the State Group and Structure Group context in great detail.

3.1 State Group

The State Group includes the first three Cognitive Dimensions:

1. Activation: Grounded (G) / Energized (N)
2. Engagement: Satisficing (S) / Elaborating (L)
3. Regulation: Internal (I) / External (E)

This gives $2^3 = 8$ possible States of Cognitive Species, listed in the following table.

Table 2: Eight States of Cognitive Species

State	Dimensions	Description
G-S-I	Grounded – Satisficing – Internal	Operates best under calm, low-stimulation conditions; relies on intuition and minimal effort to process information; self-monitors to stay on track. Often cautious and reflective, preferring stable environments.
G-S-E	Grounded – Satisficing – External	Low arousal, low-effort processor who depends on external cues, guidance, or scaffolds. Prefers instructions and feedback, and may struggle without structure.
G-L-I	Grounded – Elaborating – Internal	Low-stimulation thinker who engages deeply and thoughtfully, relying on internal monitoring. Careful, reflective, self-directed, and methodical in exploring ideas.
G-L-E	Grounded – Elaborating – External	Deep thinker at low arousal, but guided by social/environmental feedback. Prefers learning with mentors or structured collaboration, integrating external signals into reasoning.
N-S-I	Energized – Satisficing – Internal	High-arousal intuitive thinker who relies on quick heuristics but self-monitors to stay on track. Can thrive under pressure or novelty, acting fast while controlling for errors internally.
N-S-E	Energized – Satisficing – External	High-arousal, low-effort processor who depends on external guidance. Responds to stimulation but may need supervision or feedback to avoid errors.
N-L-I	Energized – Elaborating – Internal	High-energy, deeply reflective, self-directed thinker. Excels in complex, novel, or high-pressure tasks that require sustained

State	Dimensions	Description
		effort and internal regulation.
N-L-E	Energized – Elaborating – External	High-energy deep processor who integrates external cues while elaborating. Thrives with active mentorship, collaborative problem solving, or highly stimulating environments.

The States serve as the base of the beginning process of thinking and learning. The first letter tells the arousal and stimulation level. The second letter shows the depth of cognitive effort. The third letter gives the source of cognitive control.

3.2 Structure Group

The Structure Group includes the last two Cognitive Dimensions:

4. Acquisition: Observational (Ob) / Enactive (En)
5. Mapping: Building (Bd) / Reframing (Rf)

This gives $2^2 = 4$ possible Structures of Cognitive Species, listed in the following table.

Table 3: Four Structures of Cognitive Species

Structure	Dimensions	Description
En-Bd	Enactive – Building	Learn by doing and gradually improving existing structures. Prefers hands-on experimentation with clear, stepwise problem-solving. Knowledge is grounded through action and refined systematically.
En-Rf	Enactive – Reframing	Learn by doing, but frequently questions assumptions and restructures understanding. Prefers exploratory action that challenges current frameworks and seeks novel insights through experimentation.
Ob-Bd	Observational – Building	Learn by watching and modeling others, then improving existing structures. Prefers guided observation, careful analysis of examples, and applying lessons to refine practical solutions.
Ob-Rf	Observational – Reframing	Learn by observing and using insights to reframe understanding. Prefers watching patterns, questioning assumptions, and mentally reorganizing knowledge to generate new perspectives.

The structures serve as the pillar on how knowledge, behaviors, skills, or values are organized in the middle of the learning process. The first code tells the tendency of an individual's grounding based on action or observation. The second code shows the tendency of a learner to preserve or transform the structure.

3.3 Grand Classification of Cognitive Species

Now we are ready to represent all 32 Cognitive Species along with their State Groups (CD 1–3) and Structure Groups (CD 1–4). We omit the descriptions of each species and focus solely on introducing the naming convention for simplicity. The title for each species is chosen arbitrarily and should be prone to future changes.

Again, note that this is a mere classification based on the binary extrema of each dimension that describe tendencies, not fixed traits. They do not imply learning effectiveness through “matching”. Effective learning often involves movement across species depending on task demands, context, and expertise.

Table 4: Classification of Cognitive Species

No.	State Group	Structure Group	Cognitive Species
1.	G-S-I	En-Bd	Type 1A
2.		En-Rf	Type 1B
3.		Ob-Bd	Type 1C
4.		Ob-Rf	Type 1D
5.	G-S-E	En-Bd	Type 2A
6.		En-Rf	Type 2B
7.		Ob-Bd	Type 2C
8.		Ob-Rf	Type 2D
9.	G-L-I	En-Bd	Type 3A
10.		En-Rf	Type 3B
11.		Ob-Bd	Type 3C
12.		Ob-Rf	Type 3D
13.	G-L-E	En-Bd	Type 4A
14.		En-Rf	Type 4B
15.		Ob-Bd	Type 4C
16.		Ob-Rf	Type 4D
17.	N-S-I	En-Bd	Type 5A
18.		En-Rf	Type 5B
19.		Ob-Bd	Type 5C
20.		Ob-Rf	Type 5D
21.	N-S-E	En-Bd	Type 6A
22.		En-Rf	Type 6B
23.		Ob-Bd	Type 6C
24.		Ob-Rf	Type 6D
25.	N-L-I	En-Bd	Type 7A
26.		En-Rf	Type 7B
27.		Ob-Bd	Type 7C
28.		Ob-Rf	Type 7D
29.	N-L-E	En-Bd	Type 8A
30.		En-Rf	Type 8B
31.		Ob-Bd	Type 8C
32.		Ob-Rf	Type 8D

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